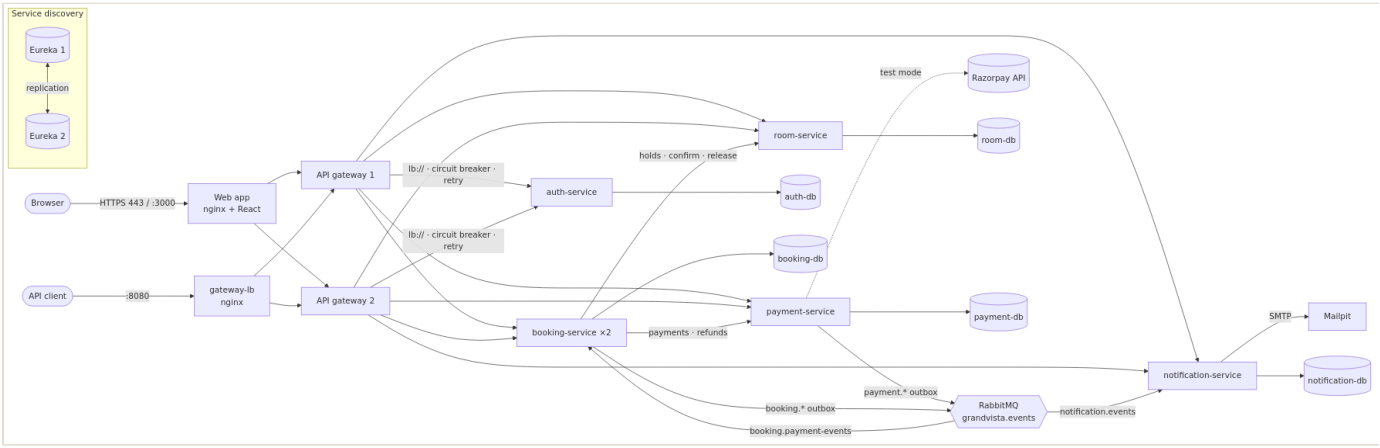
GrandVista Hospitality — Team Overview

PS016 · SOA Programming and Microservices (24SDCS03R) · K L University

Repository: https://github.com/maheswar-padi/grandvista-hospitality (public) · **Status:** CI green · **Prepared by:** PADI SESHA MAHESWARA RAO · 23 Oct 2026

1. What the project is

GrandVista is a hotel reservation platform for a **fictional** hotel, built as **Spring Boot microservices**. Customers search rooms for their dates, see a **dynamic price** (weekend, season and demand multipliers, fee and tax), reserve a room — it is **held for 15 minutes** — pay, and receive a confirmation and e-mails. Staff manage rooms, prices, bookings and payments in an admin dashboard. The core idea is a **distributed booking workflow (saga)** across independent services that never sells the same room twice, never charges without a booking, and keeps working when an instance fails. Payments use a **simulated** provider (Razorpay **test mode** is built in); no real money ever moves.



Architecture: 2 gateways behind nginx, 2 Eureka servers, 5 services with their own databases, RabbitMQ, Mailpit

2. Services and tech stack

Part	What it does
api-gateway (×2) + nginx load balancer	single entry point: routing, JWT check, CORS, rate limits, circuit breakers
discovery-server (×2, Eureka)	service registry; the two servers copy each other
auth-service	sign-up/login, RS256 JWT + refresh tokens, roles (customer, staff, admin)
room-service	rooms, availability, 15-minute holds, dynamic pricing, photos
booking-service (×2)	bookings and the saga: hold → payment → confirm, cancellations and refunds
payment-service	payment states, simulated provider / Razorpay test mode, webhooks, refunds
notification-service	turns events into e-mails (confirmation, receipt, cancellation, failed payment, expiry)
RabbitMQ · Mailpit · PostgreSQL ×5	event bus with dead-letter queues · local e-mail inbox · one database per service
frontend	React 19 + TypeScript web app for customers and admins

Stack: Java 21, Spring Boot 3.5, Spring Cloud 2025.0 (Gateway, Eureka, LoadBalancer), Resilience4j, Spring Security (JWT), JPA + Flyway, PostgreSQL 16, RabbitMQ 4.1, Thymeleaf; React 19, TypeScript, Vite, Tailwind; JUnit 5, Testcontainers, JaCoCo, Vitest, k6; Docker Compose (23 containers); GitHub Actions.

3. How to run it

Needs Docker with about **4 GB of memory**, plus `openssl`, `curl`, `jq`. From the repository folder:

```
./scripts/start.sh       # creates .env + keys if missing, builds, starts and waits (first run:
several minutes)
./scripts/e2e-smoke.sh   # 56 end-to-end checks against the running stack
```

URL	What
http://localhost:3000	web app
http://localhost:8080/api/v1 · /swagger-ui.html	API (2 gateways) · Swagger
http://localhost:8761 · :8762	Eureka 1 · Eureka 2
http://localhost:15672 · http://localhost:8025	RabbitMQ UI · Mailpit (all e-mails)

Demo logins (created automatically; every password is generated into your own `.env` and never committed):

Role	E-mail	Password in `.env`
Administrator	`admin@grandvista.local`	`GV_BOOTSTRAP_ADMIN_PASSWORD`
Customer	`guest@grandvista.local`	`GV_DEMO_CUSTOMER_PASSWORD`
Staff	`staff@grandvista.local`	`GV_DEMO_STAFF_PASSWORD`

Other useful commands: `./scripts/failover-demo.sh` (kills instances while booking), `./scripts/load-test-k6.sh`, `docker compose down` (stop, keep data). Simulated test card: **Simulated Visa – approved**.

4. Verified results (real runs)

Check	Result
GitHub Actions CI (first run 39832912584, clean runner)	**all jobs green**, ~15 min
Backend tests (unit, integration with real PostgreSQL/RabbitMQ/Mailpit)	**196 tests, 0 failures**
Code coverage (JaCoCo, CI run)	**85.9 % of lines, 65.9 % of branches**
Frontend (lint, types, tests incl. accessibility, build)	**29 tests**, all passing
End-to-end smoke test on the full stack (CI and local)	**56/56** (incl. 4 e-mails checked in Mailpit)
Database isolation (5 servers)	**40/40**
Failover: kill a gateway, a Eureka server, a booking replica (local)	run A 99/100 requests, run B 100/100; **all 8 bookings completed**
k6 load test, 20 users (local laptop)	0 % errors; search p95 135.9 ms, quote p95 60.8 ms
10 customers book the same room at the same moment	**1 accepted, 9 refused** — no double booking

5. Done vs. future work

Done: microservices with database per service · gateway + Eureka, both doubled · saga with compensations and recovery · dynamic pricing · RabbitMQ events with outboxes and dead-letter queues · e-mail notifications · JWT security and roles · simulated payments + Razorpay test-mode code · resilience (circuit breakers, retries) · React web app with admin dashboard · 225 automated tests, CI green · documentation, report draft, demo script.

Not done / future work: Razorpay never run against Razorpay's real test API (needs test keys) · no cloud deployment yet (prepared for an Oracle Cloud free VM with HTTPS) · e-mails go to Mailpit only · auth, room, payment, notification, RabbitMQ and databases run as single instances · rate limits are per gateway · report placeholders: team details, review dates, declaration/certificate pages, acknowledgement, 14 screenshots.

6. Suggested split for the reviews

Review	Focus	Show / prepare
Review 1 — design	problem, objectives, architecture, service boundaries, database per service, API design	architecture diagram, ER diagrams (`docs/database`), Swagger, `docker compose ps`, report §1–§14
Review 2 — working system	booking saga, dynamic pricing, payments, events and e-mails, security	live demo: search → book → pay → confirmation → e-mail in Mailpit; RabbitMQ queues; report §15–§20
Final Review — quality and operations	testing, high availability, CI/CD, deployment, results, limitations	failover demo, CI run and badge, coverage, k6 results, report §21–§27, all screenshots and team details filled in

Suggested ownership (adjust as a team): **member 1** — backend services and saga; **member 2** — frontend and demo walkthrough; **member 3** — testing, CI and the report. Everyone should be able to run `./scripts/start.sh` and explain the architecture diagram.

Start here: `README.md` → `docs/DEMO_SCRIPT.md` (7–10 min demo and screenshot list) → `docs/project-report/PROJECT_REPORT.md` → `docs/TRACEABILITY_MATRIX.md`.